

how far the whole measurement chain lands from what Ohm's law
predicted before the run
MEASURED 2026-09-16 · panel 2

the amplifier's own input current, against the ~1 nA it has to find — 250 times smaller

MEASURED 2026-09-15

resistance of a real tip-and-sample junction across our current-voltage sweep
MEASURED 2026-09-17 · panel 3

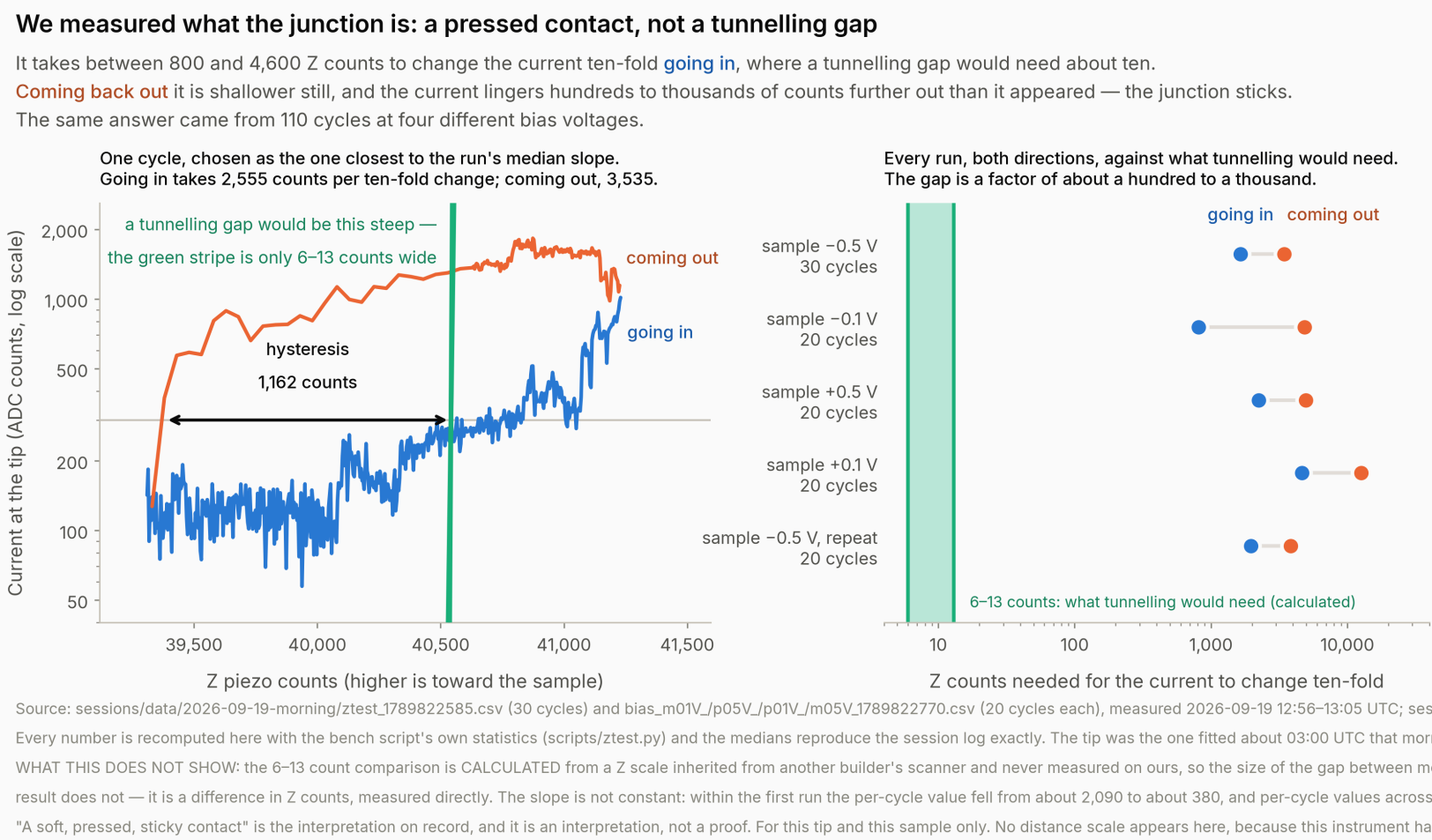
spread of one reading with the room still, then with someone stamping
two metres away
MEASURED 2026-09-19 · tip clear

Z counts the gap moved in 6.4 seconds, motor stopped, nobody touching it. **The blocker**
MEASURED 2026-09-19 · panel 5

51 recordings, 209 reproducibility tests, no demonstrable image — our own controls are how we know

2026-09-19 · panel 6

What is ours and what is not. Mechanics, controller and firmware follow Mech Panda's *red-panda-stm*; the scan head and preamplifier follow Dan Berard's home-built STM. Ours is the electronics bring-up, the software, the calibration, the controls and the fault isolation on the rest of this poster.

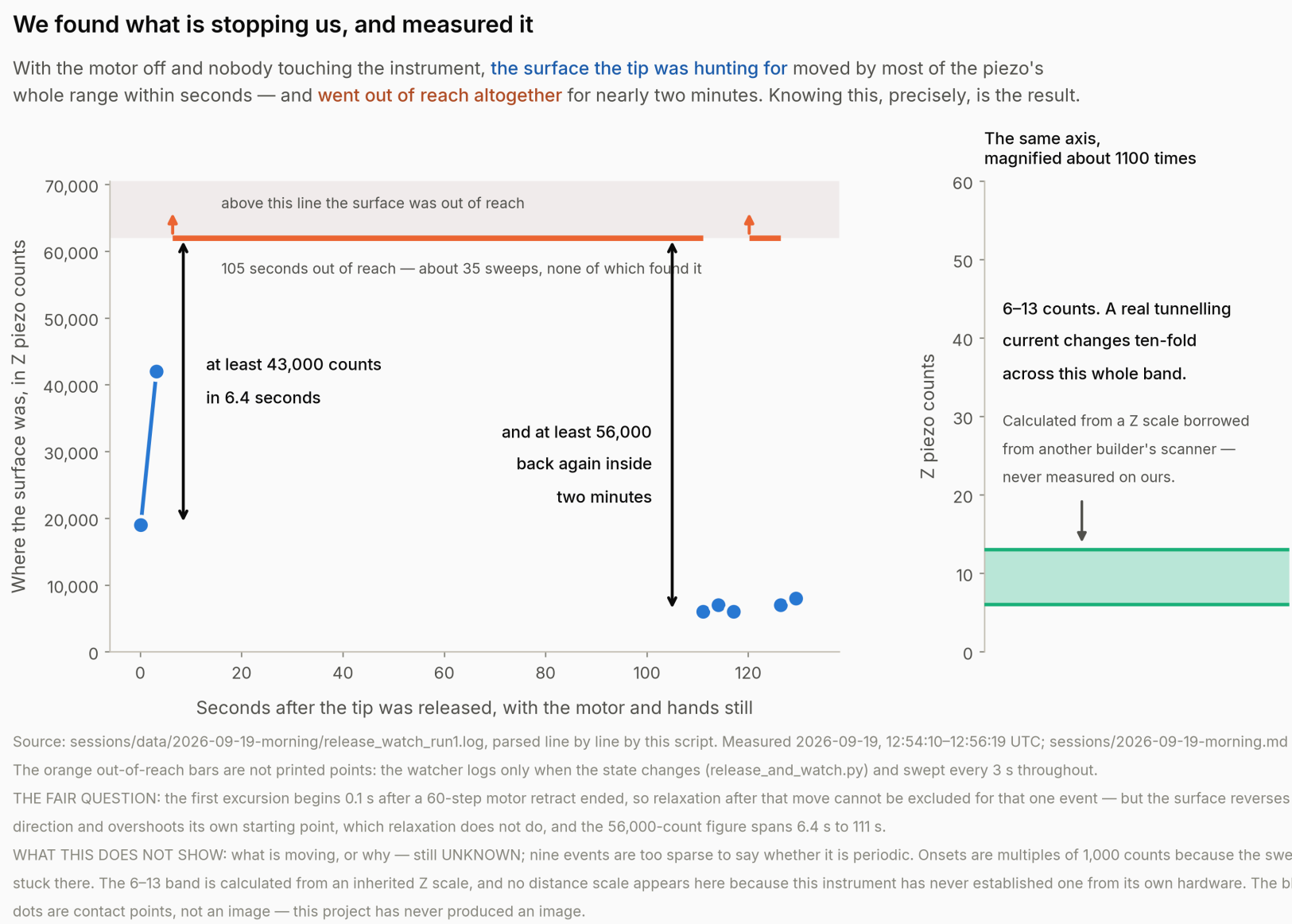


The experiment: take the plate off, lay a straightedge across the two ball ends, and see which side the tip stands on and by how much. No power, no electronics, a few minutes — and it settles our best junction retroactively.

- **No image.** Fifty-one recordings, no demonstrable image of the sample — and we know that from our own controls, not from anything turning up.
- **No demonstrated tunnelling.** We have a barrier junction; the 2026-09-19 junction was a pressed contact, measured. Our best junction's status is **undecided**, and panel 4 names the measurement that settles it.
- **No atomic resolution, and no scale bar anywhere** — every axis is in converter counts, because this instrument has never established a distance scale of its own. **No number is taken off a photograph:** every caption is marked **SAID** or **READ**.

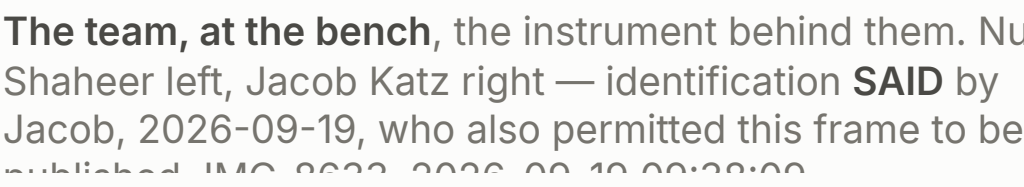


Figure 1. Measured $-3,204.8 \pm 36.5$ counts per volt against $-3,200$ predicted — agreement to 0.13 of one standard error, $R^2 = 0.9934$, 53 readings at 8 bias settings. Twelve stages had to be right at once for those points to land on that line, from the bias converter through the wire and the tip holder to the firmware and the PC tools. Source: sessions/2026-09-16-bench.md §3.4, measured 2026-09-16; re-fitted twice from the raw readings for this poster. **What it does not show:** the slope is the ratio of two nominally 100 M Ω resistors whose tolerances were never recorded, so it confirms the scale to within those tolerances rather than independently of them; and 53 readings are repeats at 8 settings, not 53 independent observations.

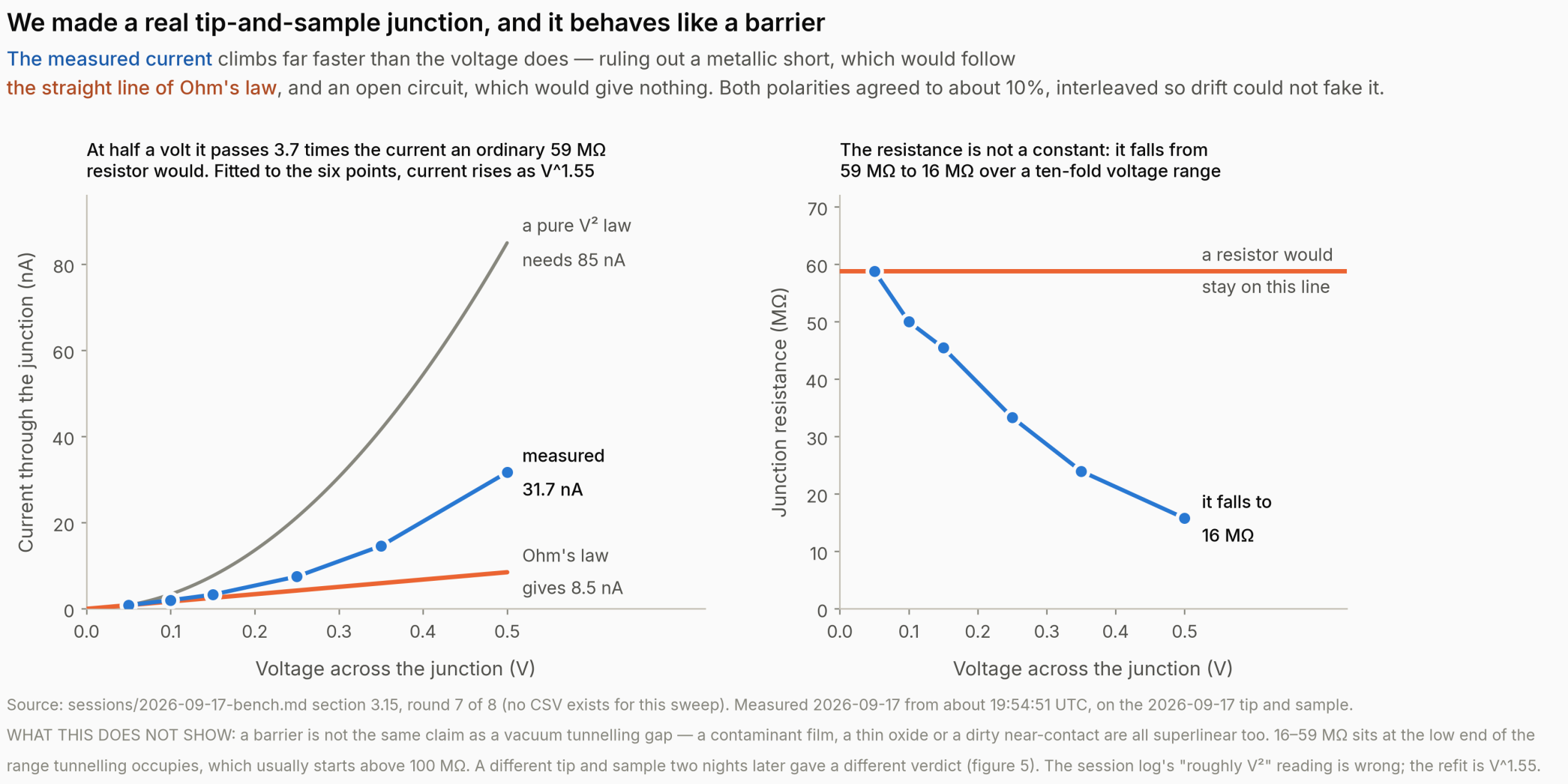


Why that ends the imaging question, in one comparison. Over the five seconds one image takes, the gap drifts **396 to 1,722 counts**; the “structure” in every candidate image we have is **29 to 458 counts**. The thing being measured is **smaller than the way the gap moves while it is being measured**. Five candidate causes, none yet tested: the leaf on its backing paper; the plate on its rubber bands and three balls; thermal motion; air currents; relaxation after a motor move. **A cardboard box over the instrument is the free first test.**

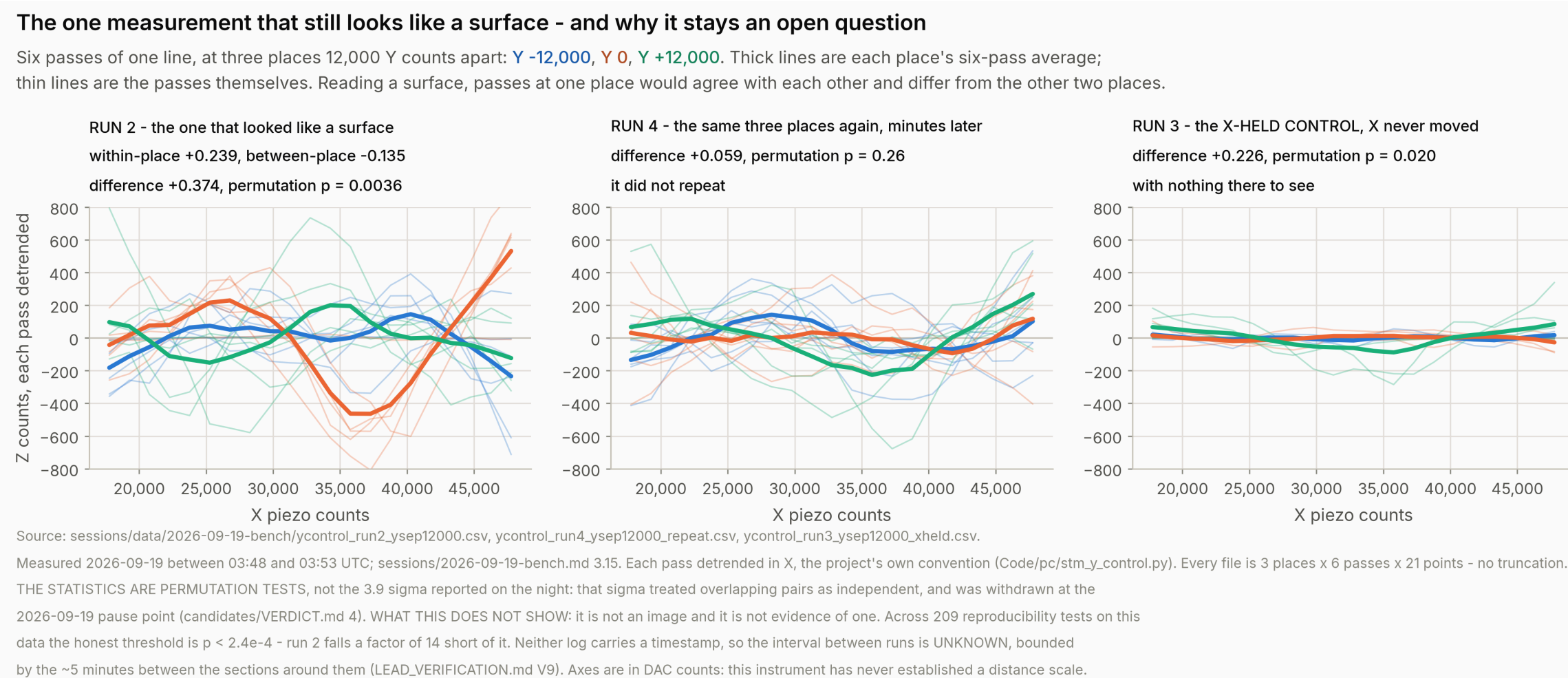
- **Measure d with a straightedge** no power, a few minutes, and it settles retroactively whether our best junction was tunnelling.
- **Prove the gap holds still before scanning anything:** a box over the instrument, a 15-minute stillness recording, a gate of 200 counts over sixty seconds.
- **A stiffer sample than gold leaf on paper, and a sharper tip.**
- **Three software fixes that cost nothing,** removing the artefact behind our most convincing false positive: drop the first three pixels of each pass, sweep X back and forth between passes, and refuse to compare two images of different length.



This is not a disclaimer. It is the result. A tunnelling microscope is normally a clean-room instrument — commercial ones sit on air tables in temperature-controlled labs — and **the blocker we found and measured is exactly the class of problem those rooms exist to remove**. Stated up front, our limitation becomes a measurement of what this environment costs you.



And the electronics are not what limits us — established with numbers rather than opinion. With no junction the chain is flat white noise at 8 to 14 counts from 2 Hz to 1.2 kHz; leakage through the tip lead is under 0.06 nA from -2 V to +2 V, more than 8 GΩ; and someone stamping two metres away moves one reading's spread from 131 to 132 pA. **The building is not getting in electrically.** MEASURED 2026-09-17 and 2026-09-19, tip clear.



What keeps it open, and the experiment that closes it. The instrument's own logs show the gap was **steadily** across these runs — the onset moved 2,200 counts, about 3% of the Z range — so “the surface moved out from under it” is not the explanation, and genuine non-reproduction is better supported. **But nothing in this project has ever measured how far the tip drifts sideways**, and that is the one ground it stays open on. **The answer: three runs with an X-held control interleaved between them, judged by a rule fixed in advance — real only if all three beat all the controls.**

At this pause point we re-derived our own published numbers from the raw files and found six of them wrong — including two statistics we had used to dismiss our own results, and a reproducible feature that turned out to be the feedback loop recovering from a jump rather than the sample. Every conclusion survived; two of the arguments for them did not. All of it is in the repository, beside the data.